Today

Agenda

- Graphical Summaries
  - Histograms, boxplots
  - Skewness, outliers
- Class exercise on sampling and descriptive statistics
- Propagation of error

Schedule

- Lab 1 is already in LearnHub. No assignment.
- Lab 2 is already in LearnHub. The assignment is due at 19:00 on Thursday, right after the lab.
- The reading assignment on Chapter 1 will be in LearnHub today.

72

Histogram

- Graphical display that gives an idea of the “shape” of the sample, indicating regions where sample points are concentrated and regions where they are sparse.
- The bars of the histogram touch each other. A space indicates that there are no observations in that interval.

73

Creating a Histogram

Choose boundary points for the class intervals. Usually these intervals are the same width.

Compute the frequencies: this is the number of observations that occur in each interval.

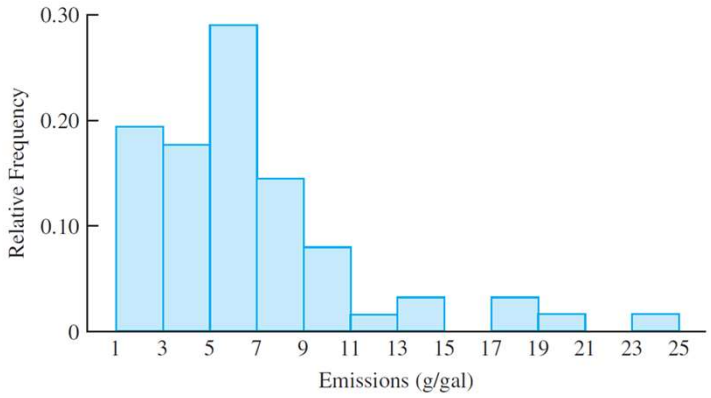
Compute the relative frequencies for each class: this is the number of observations in each interval divided by the total number of observations.

If the class intervals are the same width, then draw a rectangle for each class, whose height is equal to the frequencies or relative frequencies.

- If the class intervals are of unequal widths, the heights of the rectangles must be set equal to the densities, where **density** is the relative frequency divided by the class width.

74

Example of Histogram



75

Symmetry and Skewness [1]

A histogram is perfectly **symmetric** if its right half is a mirror image of its left half.

- For example, the distribution of birth weights of full-term infants is roughly symmetric.

Histograms that are not symmetric are referred to as **skewed**.

A histogram with a long right-hand tail is said to be **skewed to the right**, or **positively skewed.**

- For example, the distribution of household incomes is right-skewed.

A histogram with a long left-hand tail is said to be **skewed to the left**, or **negatively skewed.**

- For example, the distribution of grades on an easy test is left-skewed.

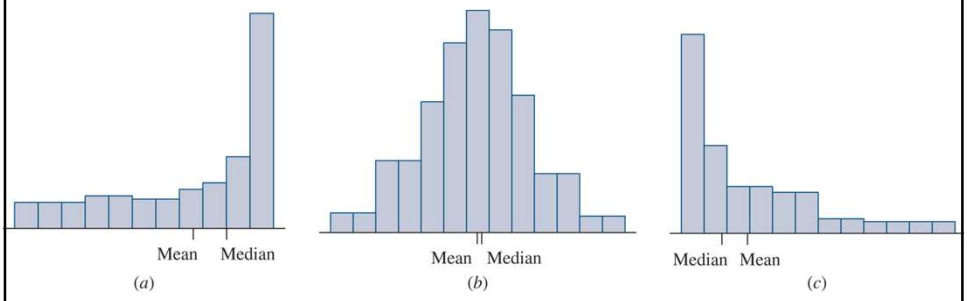
76

Symmetry and Skewness

- When a histogram is roughly **symmetric**, the mean and the median are approximately equal.
- When a histogram is **right-skewed**, the mean is greater than the median.
- When a histogram is **left-skewed**, the mean is less than the median.

77

Skewness



Skewness<0
Left skewed
(left tail is long)

Skewness≈0

Skewness>0
Right skewed
(right tail is long)

78

Unimodal and Bimodal

A histogram with only one peak is what we call **unimodal**.

If a histogram has two peaks then we say that it is **bimodal**.

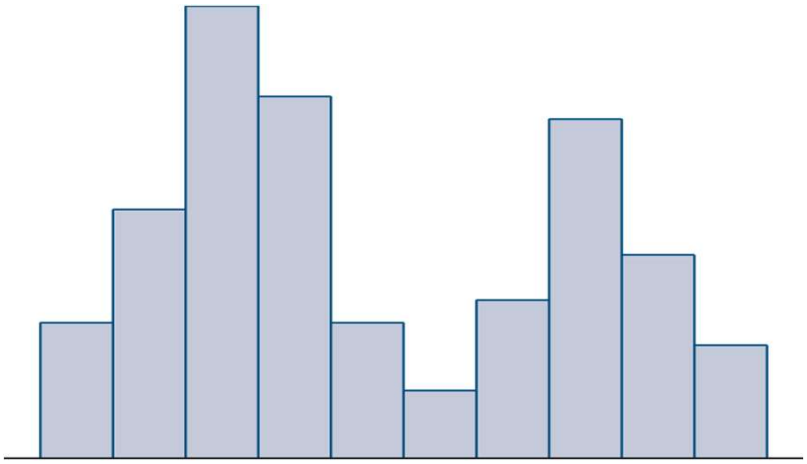
- Bimodal histograms often indicate subsamples.

If there are more than two peaks in a histogram, then it is said to be **multimodal**.

- Multimodal histograms do not often occur in practice.

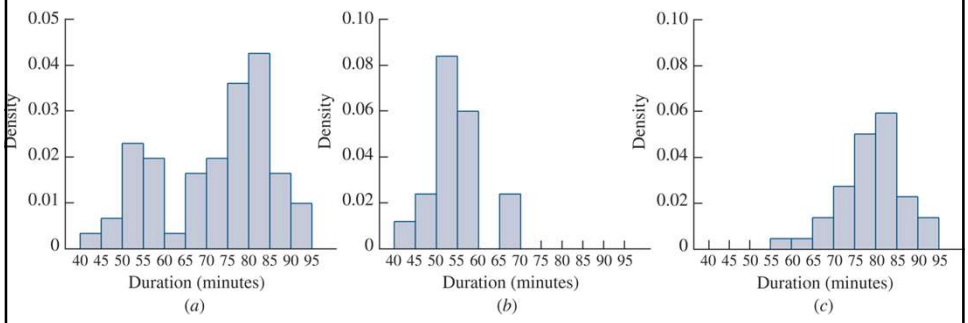
79

Bimodal data



80

Example



81

Boxplots

- A **boxplot** is a graphic that presents the median, the first and third quartiles, and any outliers present in the sample.
- The **interquartile range (IQR)** is the difference between the third quartile and the first quartile. This is the distance needed to span the middle half of the data.

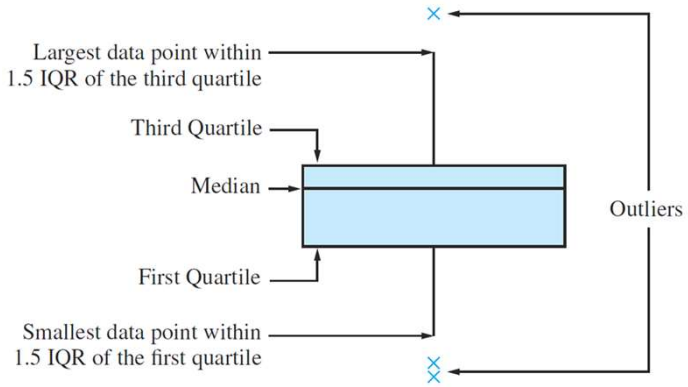
82

Creating a Boxplot

- Compute the median and the first and third quartiles of the sample. Indicate these with horizontal lines. Draw vertical lines to complete the box.
- Find the largest sample value that is no more than 1.5 IQR above the third quartile, and the smallest sample value that is no more than 1.5 IQR below the first quartile. Extend vertical lines (whiskers) from the quartile lines to these points.
- Points more than 1.5 IQR above the third quartile, or more than 1.5 IQR below the first quartile, are designated as outliers. Plot each outlier individually.

83

Anatomy of a Boxplot

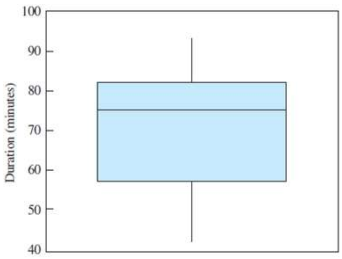


84

Example 5

- Notice there are no outliers in these data.
- Looking at the four pieces of the boxplot, we can tell that the sample values are comparatively densely packed between the median and the third quartile.
- The lower whisker is a bit longer than the upper one, indicating that the data has a slightly longer lower tail than an upper tail.
- The distance between the first quartile and the median is greater than the distance between the median and the third quartile.
- This boxplot suggests that the data are skewed to the left.

What can you say about the underlying data set?



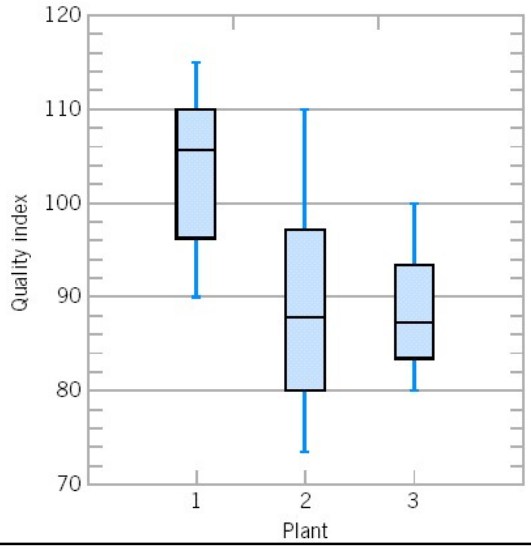
85

Comparative Boxplots

- Sometimes we want to compare more than one sample.
- We can place the boxplots of the two (or more) samples side-by-side.
- This will allow us to compare how the medians differ between samples, as well as the first and third quartile.
- It also tells us about the difference in spread between the two samples.

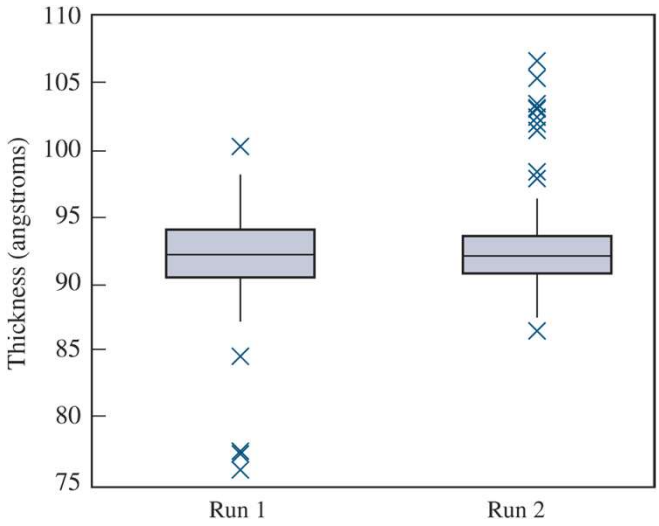
86

Comparative box plots of a quality index at three plants



87

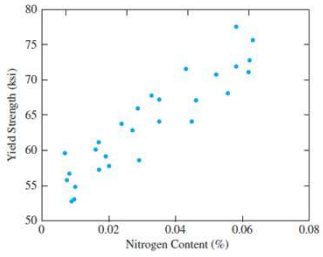
Comparative box plots of oxide layer thickness data



88

Scatterplot

- Data for which items consists of a pair of values is called **bivariate**.
- The graphical summary for bivariate data is a **scatterplot**.
- Display of a scatterplot:



89

Looking at Scatterplots

- If the dots in the scatterplot do not appear to have a general trend or pattern, then there is little evidence of a relationship between the two variables.
- If the dots on the scatterplot appear to roughly follow a straight line, then there is evidence of a linear relationship between the two variables.
- When a linear relationship exists, we may be able to use one variable to help predict the values of the other variable.

90

Summary of Chapter 1

- We discussed types of data.
- We looked at sampling methods, emphasizing simple random sampling (SRS).
- We learned about sample statistics.
- We examined graphical displays of data.

91